

Exact Return Layers and Budgeted Survivor Frontiers

Source-Relative Return Counts in Full Ordinary-Lane Dynamics

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This paper is Paper XXXVIII of the RIME program. It studies source-labelled raw first-exit incidence, charging each normalized return including the terminal occurrence. It makes no typed-transfer, recursive-settlement, or reset-bound claim.

Abstract

Problem. An unbudgeted lineage group determines which terminal placements are reachable, but need not determine their return-count layers.

Approach. We study five source lineages filling an ordinary lane in normalized circular dynamics with $n = 5g$, $\Delta = g$, and $g \geq 2$, allowing arbitrary lane-stabilizing branch restrictions and every return label. Each normalized return, including the terminal one, has unit cost.

Results. For $N \geq 1$, an exact correspondence with actual guarded paths gives $\Theta_{=N} = \mathcal{A}_a^{N-1} B_s$ and $\Theta_N = \bigcup_{\ell=0}^{N-1} \mathcal{A}_a^\ell B_s$, where $B_s = H\sigma_s$ is the forced source factor. Each product is realized without uncharged phase alignment or representative replacement. Cumulative layers cannot stall before covering K_a , so every attainable terminal placement and survivor map has a witness using at most $|K_a|/5 \leq 24$ returns. An all- $g \geq 3$ matched family has the same group, aggregate factor set, and complete unbudgeted frontier but different one-return survivor maps of equal cardinality; both frontiers coincide at two returns.

Boundary. The full-lane source is supplied, not asserted to be the complete current image reached from the initial Q . First exits and budgets are relative to these five lineages. The original-letter consequence requires a supplied normalization realization and excludes source-entry cost. No sharp constant, minimal invariant, or deterministic coset transition congruence is claimed.

Keywords. circular automata; guarded returns; source-relative budgets; positive product layers; survivor incidence

Notation Table

Symbol	Meaning
Q, E, Δ	circular carrier, normalized carrier, and lane spacing
C_j, C_*	ordinary five-point lanes and punctured four-point lane
$\lambda_{j,\kappa}$	actual placement of the five source labels
s	fixed ordinary source lane
π, σ_j	lane permutation and within-lane branch restrictions
$c, \bar{H} = \langle c \rangle$	five-position phase cycle and phase subgroup
K_a	generated five-position lineage group
$\mathcal{A}_a = \bigcup_j H\sigma_j$	aggregate factor set
$B_s = H\sigma_s$	forced initial right coset
$\Theta_{=N}, \Theta_N$	exact and at-most terminal return-count layers
μ_η	pre-collapse terminal placement of one permutation
$\mathfrak{S}_a^{\leq N}(\lambda_0; F)$	source-addressed survivor maps within budget

1 Introduction

Closure and bounded reachability read different information. In the full ordinary-lane setting, the unbudgeted reachable injections and terminal placements are classified by an embedded group $K_a \leq S_5$ [4]. A return budget asks a different question: which of those terminal permutations can be realized from a fixed source within a given number of actual returns, with injectivity preserved before the terminal collision?

The first return necessarily applies the restriction of the source lane. Later restrictions can be selected by choosing the target of the preceding return. All labels are available, so the same label also selects the current phase. The resulting products retain a source-dependent initial coset B_s as well as the aggregate factors $\mathcal{A}_a$; passing to their generated group alone forgets this information.

Theorem 3.1 is the central result. It identifies both exact and cumulative terminal layers with ordered products and constructs one actual guarded injection sequence for each claimed placement. Theorem 4.4 then gives a uniform finite return budget by counting right cosets. Theorem 5.1 shows that the initial coset is not dispensable: fixing even the aggregate set and full unbudgeted frontier does not fix the first budget layer.

The inherited uniform edge law and unbudgeted incidence classification are attributed to Paper XXXVII [4]. The new object is the return-count terminal layer and its complete survivor restriction. The local home-lane controls are recorded in Appendix A; they are not needed for the central exact layer formula or its 24-return consequence.

1.1 Related Work and Novelty Boundary

Group word lengths and positive products.

Word length depends on the generating set and the convention for inverses, not only on the generated group. Babai, Hetyei, Kantor, Lubotzky, and Seress discuss finite-group diameter with generators and their inverses [1]. Bradford uses positive-word balls and directed Cayley graphs, explicitly distinguishing them from the inverse-closed convention [2]. The present products also use positive factors; inverse steps are not free returns. No new general Cayley-diameter method is claimed.

Finite positive-generation and cardinality arguments supply the elementary saturation proof in Section 4. The contribution is not the abstract notation of product powers or the observation that a generating closure forgets lengths. It is the exact correspondence of those products, including the forced source coset, with the guarded first-exit witnesses of the declared raw system.

Almost-group and constrained synchronization.

Casas Torres studies complete reachability for automata with one defect-one letter and permutation letters, focusing on a transitive imprimitive permutation group [3]. That problem concerns images of the whole state set. Here five fixed source lineages remain distinct through every internal return and are read individually at their first strict exit.

Wolf studies synchronization under constraints on how a word traverses states [5]. Our guard instead maintains injectivity on the complete five-point placement until a terminal collision, and our cost counts normalized returns. We do not provide a general constrained-synchronization algorithm or a classification of shortest original-letter words.

Inherited geometry and scoped new claims.

Paper XXXVII supplies the uniform full-lane edge law, constructive unbudgeted reachability, and terminal survivor restriction [4]. Those results remain inherited inputs, not claims newly owned here. Theorem 3.1 supplies the return-count description with actual witnesses; its uniform bound and Theorem 5.1's symbolic source-initial-layer separation concern that new bounded observable.

The comparison is selective and does not establish exhaustive priority. Different tasks alone are not an expressivity separation or a claim of strict superiority. Likewise, $(\mathcal{A}_a, B_s)$ is a proved sufficient description of the layers, not a claimed minimal invariant. The claims are confined to the declared unit-cost return model; neither sharpness nor a typed-transfer bridge is asserted.

2 Raw System and Return-Count Convention

Retain the raw setup of Paper XXXVII version 1.0 [4]. Fix

$$n = 5g, \quad \Delta = g, \quad g \geq 2, \quad Q = \mathbb{Z}/n\mathbb{Z}, \quad E = Q \setminus \{0\}, \quad (2.1)$$

with $p(q) = q + 1$ and $\varepsilon_\Delta(0) = \Delta$, $\varepsilon_\Delta(q) = q$ for $q \neq 0$. Its unique collision pair is $K_\Delta = \{0, \Delta\}$. The ordinary lanes and punctured lane are

$$\begin{aligned} C_j &= \{j + t\Delta : t \in \mathbb{Z}/5\mathbb{Z}\}, \quad 1 \leq j < g, \\ C_* &= \{\Delta, 2\Delta, 3\Delta, 4\Delta\}. \end{aligned} \quad (2.2)$$

The branch $a \in \text{Sym}(E)$ stabilizes this lane partition. There are a permutation π of the ordinary lane indices and arbitrary $\sigma_j \in S_5 = \text{Sym}(\mathbb{Z}/5\mathbb{Z})$ such that

$$a(j + t\Delta) = \pi(j) + \sigma_j(t)\Delta. \quad (2.3)$$

The restriction to C_* is arbitrary. Fix source labels $T_0 = \{q_0, \dots, q_4\}$, a source lane $s = j_{\text{src}}$, and

$$\lambda_0 = \lambda_{s, \text{id}}, \quad \lambda_{j, \kappa}(q_i) = j + \kappa(i)\Delta. \quad (2.4)$$

These five labelled lineages form a supplied local configuration. They are not asserted to be all current packets descended from the full initial carrier Q ; first exit below is relative to this configuration.

Source indices are read modulo five. Set

$$c(t) = t + 1, \quad H = \langle c \rangle, \quad K_a = \langle c, \sigma_1, \dots, \sigma_{g-1} \rangle \leq S_5. \quad (2.5)$$

Every label $r \in Q$ is available. The normalized return is $\phi_r^{(a)} = \varepsilon_\Delta p^r a$. Composition is right-to-left; a word lists labels in execution order. The inherited edge law, Paper XXXVII Lemma 3.1 / equation (3.2), is

$$\begin{aligned} \pi(j) + r &= h + k\Delta \pmod{n}, \quad 1 \leq h < g, \\ \lambda_{j, \kappa} &\xrightarrow{r} \lambda_{h, c^k \sigma_j \kappa}. \end{aligned} \quad (2.6)$$

This edge is uniformly enabled on every injection filling C_j . If $\pi(j) + r \equiv 0 \pmod{g}$, the corresponding occurrence is strict: its pre-collapse support is $\widehat{C}_* = \{0, \Delta, 2\Delta, 3\Delta, 4\Delta\}$. The punctured lane cannot receive five distinct lineages internally.

A terminal witness has an actual guarded prefix followed by one strict return. Its return count N_{ret} includes that last occurrence. Thus an N -return witness has $N - 1$ internal returns and one terminal return. An empty internal prefix is allowed; zero returns cannot realize a terminal event.

Write

$$\mu_\eta(q_i) = \eta(i)\Delta, \quad \eta \in K_a. \quad (2.7)$$

Paper XXXVII, Theorem 5.1 [4], identifies these with all actual unbudgeted pre-collapse terminal placements. Define $\Theta_{=N}(a, s)$ by actual witnesses using exactly N returns, and define $\Theta_N(a, s)$ by actual witnesses using at most N returns:

$$\begin{aligned}\Theta_{=N} &= \{\eta \in K_a : \mu_\eta \text{ has an actual } N\text{-return witness}\}, \\ \Theta_N &= \bigcup_{t=1}^N \Theta_{=t}, \quad N \geq 1.\end{aligned}\tag{2.8}$$

The prefix must remain injective at every internal return. The final return is therefore the first strict rank drop.

For subsets of a group, $XY = \{xy : x \in X, y \in Y\}$, $X^0 = \{1\}$, and $X^{\ell+1} = XX^\ell$. No inverse is treated as a free generator or a free return.

3 Exact Return Layers and Same-Witness Realization

Define the existing aggregate factor set and forced source coset by

$$\mathcal{A}_a = \bigcup_{1 \leq j < g} H\sigma_j, \quad B_s = H\sigma_s.\tag{3.1}$$

Theorem 3.1.

For every $N \geq 1$,

$$\boxed{\begin{aligned}\Theta_{=N}(a, s) &= \mathcal{A}_a^{N-1} B_s, \\ \Theta_N(a, s) &= \bigcup_{\ell=0}^{N-1} \mathcal{A}_a^\ell B_s.\end{aligned}}\tag{3.2}$$

For every element, the construction produces one actual witness with the stated return count, retaining the same concrete representative from step to step.

Proof. actual paths imply products. Consider an actual N -return witness. Put $j_0 = s$ and $\kappa_0 = \text{id}$. For internal step t , let its ordinary target be C_{j_t} and write its phase choice as $k_t \in \mathbb{Z}/5\mathbb{Z}$. Equation (2.6) gives

$$\kappa_t = c^{k_t} \sigma_{j_{t-1}} \kappa_{t-1}.\tag{3.3}$$

The last step produces $\eta = c^{k_N} \sigma_{j_{N-1}} \kappa_{N-1}$. Consequently,

$$\eta = (c^{k_N} \sigma_{j_{N-1}}) \cdots (c^{k_2} \sigma_{j_1}) (c^{k_1} \sigma_s).\tag{3.4}$$

The first chronological factor, on the right, belongs to B_s . Each later factor belongs to $\mathcal{A}_a$. This proves the forward inclusion for the exact layer.

products imply actual paths. Conversely, choose a product as in (3.4), with $j_0 = s$, ordinary $j_1, \dots, j_{N-1}$, and phases $k_1, \dots, k_N$. For $1 \leq t < N$, choose

$$r_t = j_t - \pi(j_{t-1}) + k_t \Delta \pmod{n}.\tag{3.5}$$

Starting at λ_0 , this one return both applies the desired factor and places the resulting injection in the ordinary lane needed for the next factor. It is uniformly enabled by (2.6). The actual successor determines the next concrete input; no representative is exchanged in a decoder fiber.

For the last occurrence choose

$$u = -\pi(j_{N-1}) + k_N \Delta \pmod{n}.\tag{3.6}$$

It gives the pre-collapse placement μ_η and has support $\widehat{C}_*$, hence is strict. The preceding $N - 1$ returns were injective, so this is the first rank drop. When $N = 1$, the construction has no internal step and uses $j_0 = s$ directly.

Taking the union over $1 \leq t \leq N$ proves the cumulative formula. The phase choice in each factor is supplied by that return's label; no separate phase path or uncharged relocation has been inserted. $\square$

In particular,

$$\Theta_1 = B_s = H\sigma_s. \quad (3.7)$$

The forced source factor cannot be replaced by H or discarded. The result preserves the fixed source and desired terminal placement, not an independently preassigned terminal exponent, prefix, or historical witness.

For a source pair F , let $I_F = \{i : q_i \in F\}$ and $R_0 = T_0 \setminus F$. Define $\mathfrak{S}_a^{\leq N}(\lambda_0; F)$ from actual terminal witnesses whose return count is at most N , just as the unbudgeted frontier is defined in Paper XXXVII.

Proposition 3.2.

$$\boxed{\mathfrak{S}_a^{\leq N}(\lambda_0; F) = \{q_i \mapsto \eta(i)\Delta \mid (q_i \in R_0) : \eta \in \Theta_N, \quad \eta(I_F) = \{0, 1\}\}.} \quad (3.8)$$

The same statement for exactly N returns uses $\Theta_{=N}$.

Proof. Every actual budgeted terminal placement has permutation $\eta \in \Theta_N$ by Theorem 3.1. It fuses F precisely when $\eta(I_F) = \{0, 1\}$. Its three survivor reads are the indicated restriction. Conversely, Theorem 3.1 constructs one actual witness for every such η , within the stated budget. That same witness supplies both the fused pair and its survivor map. $\square$

Thus $(\mathcal{A}_a, B_s)$ determines the return-count terminal layers. For fixed n, Δ , and source labels, equal aggregate sets and initial cosets determine equal budgeted survivor frontiers. This is a sufficiency statement, not minimality, a converse, or equality of labelled path languages. The lane permutation is still used to construct the actual labels in (3.5)–(3.6).

4 Uniform Finite Return-Budget Realization

Lemma 4.1.

The group and the positive monoid generated by $\mathcal{A}_a$ both equal K_a .

Proof. Every element of $\mathcal{A}_a$ belongs to K_a . For each j , the aggregate contains both σ_j and $c\sigma_j$. Its generated group therefore contains all σ_j and $c = (c\sigma_j)\sigma_j^{-1}$, hence equals K_a . Each inverse of an aggregate element is a nonnegative power of that element because the group is finite. Positive generation consequently has the same closure. This is an algebraic closure argument, not a permission to execute a free inverse return. $\square$

Lemma 4.2.

For $N \geq 1$,

$$\Theta_{N+1} = \Theta_N \cup \mathcal{A}_a \Theta_N, \quad H\Theta_N = \Theta_N. \quad (4.1)$$

Each Θ_N is a nonempty union of five-element right cosets $H\kappa$.

Proof. The recurrence follows by distributing set multiplication over the finite union in (3.2). The initial coset is left H -invariant, as is $\mathcal{A}_a = \bigcup_j H\sigma_j$. Each term in that union is therefore left H -invariant. Since H has order five, its left action on the group is free and its orbits are the five-element right cosets $H\kappa$. The initial layer has five elements and every cumulative layer contains it. $\square$

The recurrence concerns sets of attainable terminal permutations. It does not append an internal step to a completed strict exit. To realize a new product, Theorem 3.1 constructs a new path, choosing ordinary targets for all factors except the last. In particular, it can retarget what would have been a shorter witness's final factor.

Nor does right-coset closure make left multiplication by each aggregate element a deterministic map on right cosets. The normalizer condition for that different assertion remains the one in Paper XXXVII.

Lemma 4.3.

If $\Theta_{N+1} = \Theta_N$, then $\Theta_N = K_a$. Consequently,

$$\Theta_N \neq K_a \implies |\Theta_{N+1}| \geq |\Theta_N| + 5. \quad (4.2)$$

Proof. Equality and (4.1) imply $x\Theta_N \subseteq \Theta_N$ for every $x \in \mathcal{A}_a$. Left multiplication is a bijection of the finite group, so $|x\Theta_N| = |\Theta_N|$ and the inclusion is equality. It follows that Θ_N is invariant under each such left multiplication and its inverse, and hence under the generated group K_a .

Choose $y \in \Theta_N$. For every $z \in K_a$, the element zy^{-1} belongs to K_a , so invariance gives $z = (zy^{-1})y \in \Theta_N$. The reverse inclusion already holds, proving $\Theta_N = K_a$.

If the cumulative layer is proper, it must therefore grow strictly. Both layers are unions of five-element right cosets, so strict growth adds at least five elements. $\square$

Theorem 4.4.

Put $q = |K_a|/5$, an integer because $H \leq K_a$. Then

$$\boxed{\Theta_q = K_a, \quad q \leq 24.} \quad (4.3)$$

Every actual attainable terminal placement has one guarded witness whose first strict return is terminal and whose total return count is at most q .

Proof. K_a has exactly q right cosets of H . The initial layer contains one of them. By Lemma 4.3, each successive cumulative layer either has already reached K_a or adds at least one new coset. After at most $q - 1$ expansions, the layer is full. This includes $q = 1$, where the initial coset is already all of K_a . Finally, $|K_a| \leq 120$ gives $q \leq 24$.

Every $\eta \in K_a$ belongs to the cumulative layer Θ_q . Theorem 3.1 supplies the actual guarded prefix and terminal occurrence realizing μ_η within this budget. Paper XXXVII's terminal classification identifies these with all unbudgeted attainable placements. $\square$

By Proposition 3.2, the same bound realizes every unbudgeted survivor map, with its fused pair read from the same terminal permutation. Equivalently,

$$\mathfrak{S}_a^{\leq q}(\lambda_0; F) = \mathfrak{S}_a(\lambda_0; F) \quad \text{for every source pair } F. \quad (4.4)$$

The bound concerns cumulative layers, not $\Theta_{=q}$. Exact layers need not be monotone: if every local restriction is odd, all aggregate elements are odd because a five-cycle is even, and every exact N -return terminal permutation has parity $(-1)^N$. No assertion that every placement can be padded to exactly 24 returns, or that 24 is a sharp constant, is made.

5 An All-g Budget-Frontier Non-Descent Family

Theorem 5.1.

Fix $g \geq 3$, the same source $\lambda_0(q_i) = 1 + i\Delta$, $\pi = \text{id}$, and identity on the punctured lane. Let $\tau(t) = -t \pmod{5}$ and define

$$\begin{aligned} a^{(2)} : \quad & \sigma_2 = \tau, \quad \sigma_j = \text{id} \ (j \neq 2), \\ a^{(1)} : \quad & \sigma_1 = \tau, \quad \sigma_j = \text{id} \ (j \neq 1). \end{aligned} \tag{5.1}$$

Both branches have the same $K_a = D_{10}$, complete reachable injection set, terminal-placement set, and unbudgeted survivor frontier. They also have the same unsigned labelled lane-exit graph. For each consecutive source pair $F_i = \{q_i, q_{i+1}\}$,

$$\begin{aligned} \mathfrak{S}_{a^{(2)}}^{\leq 1}(\lambda_0; F_i) &= \{s_i^+\}, \\ \mathfrak{S}_{a^{(1)}}^{\leq 1}(\lambda_0; F_i) &= \{s_i^-\}, \end{aligned} \quad s_i^+ \neq s_i^-, \tag{5.2}$$

where, in the ordered survivor labels $(q_{i+2}, q_{i+3}, q_{i+4})$,

$$s_i^+ = (2\Delta, 3\Delta, 4\Delta), \quad s_i^- = (4\Delta, 3\Delta, 2\Delta). \tag{5.3}$$

Both one-return sets have cardinality one. At two returns the complete sets agree:

$$\mathfrak{S}_{a^{(2)}}^{\leq 2}(\lambda_0; F_i) = \mathfrak{S}_{a^{(1)}}^{\leq 2}(\lambda_0; F_i) = \{s_i^+, s_i^-\}. \tag{5.4}$$

Proof. Both branches contain an identity restriction and a reflection restriction, so

$$K_a = \langle c, \tau \rangle = H \sqcup H\tau = D_{10}, \quad \mathcal{A}_a = D_{10}. \tag{5.5}$$

Paper XXXVII, Theorems 4.1, 5.1, and 5.2, now gives identical unbudgeted injections, terminal placements, and survivor sets. The unsigned labelled graph depends on π and the lane partition, not on the local restrictions, so it agrees as well.

The source is C_1 . By (3.7), its forced first layers are

$$\Theta_1(a^{(2)}, 1) = H, \quad \Theta_1(a^{(1)}, 1) = H\tau. \tag{5.6}$$

For F_i , the unique rotation sending its two indices to $\{0, 1\}$ is $\eta_i^+(t) = t - i$. The unique reflection doing so is $\eta_i^-(t) = i + 1 - t$. Their restrictions to the three ordered survivor labels give exactly (5.3). Proposition 3.2 supplies actual single-return witnesses for both. The coordinates 2Δ and 4Δ are distinct in Q , so the survivor maps differ even though their counts are equal.

For either branch, $\mathcal{A}_a = D_{10}$ and Θ_1 is a nonempty coset in that group. Thus $\mathcal{A}_a \Theta_1 = D_{10}$ and (4.1) gives $\Theta_2 = D_{10}$. Restriction yields (5.4). Nonconsecutive pairs have empty survivor sets throughout, since dihedral permutations preserve five-cycle adjacency. $\square$

The family keeps even the aggregate factor set fixed. What differs is the forced initial coset B_s . Therefore

same K_a, same complete reachable injections, same unbudgeted terminal and survivor sets $\not\Rightarrow$ same return-budgeted survivor frontier.

(5.7)

This is a full source-addressed set separation, not a count separation, a search failure, or a universal minimality statement.

6 Conditional Original-Letter Consequence

Corollary 6.1.

Suppose an existing normalization realization binds the fixed normalized source to a concrete raw section source. Fix its parameter $t_0 \in Q$ and suppose it represents label r by the chronological original block

$$p^{[t_0+r]_n}d, \quad 0 \leq [t_0 + r]_n < n. \quad (6.1)$$

Assume this same fixed realization covers the internal states and terminal occurrences of the witnesses constructed in Theorem 4.4, transporting their guards and strict events on the corresponding source-labelled raw invocation. Then every normalized attainable terminal placement has a corresponding raw strict-exit witness, from that fixed section source, with

$$|w_{\text{raw}}| \leq n |K_a|/5 \leq 24n. \quad (6.2)$$

Proof. Theorem 4.4 supplies at most $|K_a|/5$ returns on one actual witness. Expand those returns through the supplied blockwise realization. Each block has at most n original letters. The assumed guard/event transport makes every nonterminal block rank-preserving and the last block strict. The pure p prefixes preserve rank, so no earlier strict drop is introduced. Concatenating these same-source blocks gives (6.2). $\square$

The fixed normalization realization is an additional hypothesis. No source-specific realization theorem is invoked here to discharge it; the abstract normalized model alone does not supply that binding. Thus (6.2) is a proved conditional consequence, not an independent original-letter length result.

This paper does not supply an original normalization realization for an unspecified source. This conditional bound excludes entry cost, does not preserve a chosen old word or exponent, and is not a shortest-word classification. Labels with different $[t_0 + r]_n$ have different raw costs, although every phase choice costs one normalized return.

7 Scope and Open Boundaries

The central result is the exact return-count layer formula with its actual guarded realization. Uniform finite-budget saturation and the symbolic initial-layer separation are its quantitative and negative consequences; neither requires a subgroup census or a shortest-word classification.

The source assumption is not an entry-supply conclusion. Under the frozen lane-stabilizing actions, reduction modulo g intertwines each return with a permutation of residue classes: a permutes lanes, p^r translates residues, and ε_g preserves them. Thus a support occupying more than one residue class cannot enter a single full ordinary lane under these returns. In particular, the full normalized carrier E cannot supply this entrance. This does not change the source-relative first-exit or budget results.

The all- g statements are proved under the inherited raw full-lane assumptions. Bounded replay checks only their declared finite instances. The partial Lean spine covers selected algebraic steps, not the concrete guarded realization, and does not import Paper XXXVII's formalization.

The three notions remain distinct:

exact return-count terminal layers $\neq$ shortest original-word classification $\neq$ recursive credit budget or settlement.	(7.1)
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Partial occupancy, restricted return-label menus, typed source supply, target-blind authorization, selector identity, transfer membership, handoff, POS, recursive compatibility, and reset bounds are not supplied. These results concern the declared full-lane return-count observable only; the unbudgeted classification is an inherited input, not a budget theorem attributed to the earlier published version.

8 Computational Artifacts

The paper-owned evidence package is under `experiments/paper38/` in the [RIME repository](#). Paths below are relative to this directory.

Surface	Location	Role
Scope	<code>audit_scope.py</code>	fixed finite catalogue and budget range
Control	<code>return_budget_audit.py,</code> <code>results/</code>	return layers and actual witness checks
Lean	<code>lean/</code>	partial algebraic spine and source-addressed coverage
Validation	<code>validation/</code>	separately implemented coordinate replay and source binding
Evidence	<code>development-manifest.json</code>	exact selected source closure
Release	<code>release-manifest.json</code>	manuscript, PDF, and environment binding
Receipt	<code>results/</code>	downstream local closure verification

The controls exhaust all 120 local restrictions at $g = 2$ with the declared punctured identity, and use a selected catalogue at $g = 3, 4$. Every ordinary source lane and exact return counts $1, \dots, 26$ are checked. The symbolic matched family is additionally replayed at $g = 3, 4, 5$. These are bounded consistency controls, not proofs of the all- g results or independent Computational Certificates. The validator reconstructs the full labelled injection layers and survivor maps rather than trusting stored counts.

The partial Lean algebraic spine checks positive product layers, cumulative phase-block growth and saturation, the algebraic 24-factor consequence, and the finite D_{10} initial-layer control. It is not a complete formalization of the all- g geometric results: concrete guards, actual label-word realization, physical survivor embedding, and the original-letter corollary remain outside its checked scope.

9 Claim Status and Boundary

Claim	Status
Theorem 3.1 and Proposition 3.2: exact layers and survivor restriction	proved with same-witness realization
Lemmas 4.1–4.3 and Theorem 4.4: cumulative saturation	proved; at most 24 normalized returns
Theorem 5.1: matched initial-layer separation	proved for all $g \geq 3$ in the declared family
Corollary 6.1: original-letter bound	Proved conditional corollary; the fixed normalization realization is not established here
Lemma A.1 and Proposition A.2: home-lane controls	Proved auxiliary lemma and proposition; not premises of Theorem 3.1 or 4.4
Retained finite replay	Bounded consistency control on the declared finite catalogue; not an all- g proof
Lean algebraic spine	Partial formalization; concrete guards and survivor realization remain unformalized

The proved mathematical statements have theorem-level evidence under their declared hypotheses. Partial formalization describes proof coverage, not a separate evidence level or an enlargement of those hypotheses.

10 Conclusion

The fixed source enters the terminal layer law through $B_s = H\sigma_s$. All later factors come from $\mathcal{A}_a$, and choosing the next lane and phase within each return realizes the products on one actual guarded sequence. The cumulative layers saturate within $|K_a|/5 \leq 24$ returns without a normalizer assumption or a deterministic coset update.

The matched family fixes $\mathcal{A}_a$, K_a , and all unbudgeted incidence sets while changing the first-layer survivor maps. Its second layer recovers the common complete frontier. Thus the generating closure is adequate for unbudgeted existence but omits an initial condition that can matter for return-budgeted incidence. The conditional original-letter bound does not identify a shortest-word metric or a recursive settlement law.

Appendix A Local Home-Lane Controls

Lemma A.1.

Let $\alpha_j = j - \pi(j) \pmod{n}$ and $m_j = \text{ord}(\sigma_j)$. Actual uniformly guarded words realize phase and pure relocation with at most six returns. At any ordinary home lane, an actual loop realizes any σ_j with at most thirteen returns.

Proof. By the cycle types of a permutation on five points, $m_j \in \{1, 2, 3, 4, 5, 6\}$. Label α_j is an internal self-return on C_j and applies σ_j to its current permutation. The word

$$\underbrace{\alpha_j; \dots; \alpha_j}_{m_j-1 \text{ returns}}; (\alpha_j + \Delta) \quad (\text{A.1})$$

acts as $(c\sigma_j)\sigma_j^{m_j-1} = c$, staying in C_j . Replacing its final label by $h - \pi(j)$ gives

$$T_{j \rightarrow h} = \underbrace{\alpha_j; \dots; \alpha_j}_{m_j-1 \text{ returns}}; (h - \pi(j)). \quad (\text{A.2})$$

The total within-lane permutation is $\sigma_j \sigma_j^{m_j-1} = \text{id}$, and the final support is C_h . Both words have length $m_j \leq 6$; when $m_j = 1$, the repeated block is empty. All intermediate steps stay in an ordinary lane, and the last step has the explicitly chosen ordinary target. Equation (2.6) proves legality on the same representative throughout.

At a home lane C_h , execute

$$T_{h \rightarrow j}; 0; T_{\pi(j) \rightarrow h}. \quad (\text{A.3})$$

The first and last blocks preserve indices. The zero return applies σ_j while moving C_j to $C_{\pi(j)}$. The loop therefore applies σ_j at C_h at cost at most $6 + 1 + 6 = 13$. $\square$

Proposition A.2.

Every $\eta \in K_a$ has an actual terminal witness with

$$N_{\text{ret}} \leq 13(|K_a| - 1) + 7 \leq 1554. \quad (\text{A.4})$$

Proof. In a finite group, the positive monoid generated by $S = \{c, \sigma_1, \dots, \sigma_{g-1}\}$ is the group it generates: an inverse of each generator is a nonnegative power of that generator. Thus every element of K_a is reachable from the identity in the positive Cayley graph. A shortest such path repeats no vertex and has at most $|K_a| - 1$ edges.

Choose an ordinary incoming lane C_j . Relocate there at cost at most six. Use Lemma A.1 to realize a positive expression for $\kappa = \sigma_j^{-1}\eta$ at that home lane, at cost at most $13(|K_a| - 1)$. Finally take $u = -\pi(j)$. The inherited terminal law gives $p^u a \lambda_{j,\kappa} = \mu_\eta$, on this same actual prefix and terminal occurrence. Adding the last return gives (A.4), since $|K_a| \leq |S_5| = 120$. $\square$

This construction controls permutations at a prescribed home lane without using the global order of a . It is not used to prove the exact layer formula or Theorem 4.4.

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